FIITJEE Students From All Programs have bagged 34 in Top 100, 66 in Top 200 and 174 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2014:
2521 FIITJEE Students from Classroom / Integrated School Programs & 3579 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2014.

ALL INDIA TEST SERIES

FIITJEE **JEE (Advanced), 2015**

CONCEPT RECAPITULATION TEST - II

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 to 10)** contains 10 multiple choice questions which have only one correct answer. Each question carries **+2 marks** for correct answer. There is no negative marking.

Section-A (11 to 15) contains 5 multiple choice questions which have more than one correct answer. Each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

(ii) **Section-C (01 to 05)** contains 5 Numerical based questions with answers as numerical value and each question carries **+4 marks** for correct answer and **– 1 mark** for wrong answer.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

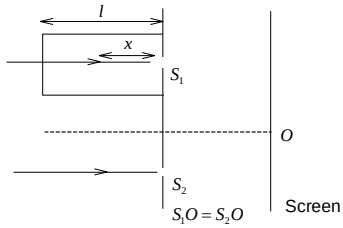
PART – I

SECTION – A

Single Correct Choice Type

This section contains **10 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D) out of which **ONLY ONE option is correct**.

1. In the figure shown, a parallel beam of light is incident on the plane of the slits of a Young's double slit experiment. Light incident on the slit, S, passes through a medium of variable refractive index $\mu = 1 + ax$ (where x is distance from the plane of slits as shown) upto a distance 'l' before falling on S_1. Rest of the space is filled with air. If at 'O' a minima is formed, then the minimum value of the positive constant a is:



(A) $\frac{\lambda}{l}$ (B) $\frac{\lambda}{l^2}$

(C) $\frac{l^2}{\lambda}$ (D) $\frac{l^2}{2\lambda}$

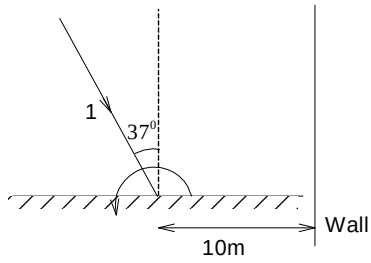
2. The fundamental frequency of a sonometer wire of length l is f_0. A bridge is now introduced at a distance Δl from the centre of the wire $(\Delta l << l)$. The number of beats heard if both sides of the bridges are set into vibration in their fundamental modes are

(A) $\frac{8f_0\Delta l}{l}$ (B) $\frac{f_0\Delta l}{l}$

(C) $\frac{2f_0\Delta l}{l}$ (D) $\frac{4f_0\Delta l}{l}$

Space for Rough work

3. A light ray 1 is incident on a plane mirror M. The mirror is rotated in the direction as shown in fig. by an arrow at frequency $\frac{9}{\pi}$rps . The light reflected by the mirror is received on the wall W at a distance 10 m from the axis of rotation. When the angle of incidence is 37^0, the speed of the spot on the wall is:



(A) $10\,m/s$ (B) $1000\,m/s$

(C) $500\,m/s$ (D) $100\,m/s$

4. The photon radiated from hydrogen corresponding to 2nd line of Lyman series is absorbed by a hydrogen like atom x in 2nd excited state. As a result the hydrogen like atom 'x' makes a transition to nth orbit. Then

(A) $x = He^{+}, n = 4$ (B) $x = Li^{++}, n = 6$

(C) $x = He^{+}, n = 6$ (D) $x = Li^{++}, n = 9$

5. If a tuning fork of frequency (f_0) 340 Hz tolerance $\pm 1\%$ is used in resonance column method . The first and the second resonance are measured at $l_1 = 24.0$ cm and $l_2 = 74.0$ cm. The maximum permissible error in speed of sound is:

(A) 1.2 % (B) 1.8 %

(C) 1 % (D) 0.8 %

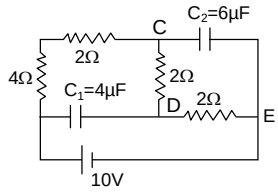
6. Axis of a solid cylinder of infinite length and radius R lies along y – axis it carries a uniformly distributed current 'I' along + y direction. Magnetic field at a point $\left(\frac{R}{2}, y, \frac{R}{2}\right)$ is

(A) $\frac{\mu_0 I}{4\pi R}\left(\hat{i} - \hat{k}\right)$ (B) $\frac{\mu_0 I}{2\pi R}\left(\hat{j} - \hat{k}\right)$

(C) $\frac{\mu_0 I}{4\pi R}\hat{j}$ (D) $\frac{\mu_0 I}{4\pi R}\left(\hat{i} + \hat{k}\right)$

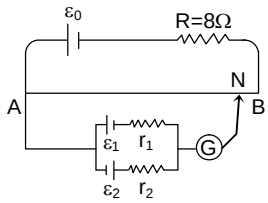
Space for Rough work

7. Find the potential difference across C_2 : When steady state reached.



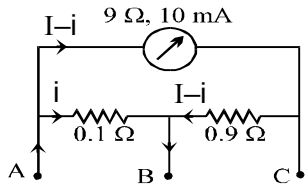
(A) 6 V (B) 4 V
(C) Zero (D) 2 V

8. A battery of emf $\varepsilon_0 = 12V$ is connected across a 4m long uniform wire having resistance 4Ω/m. The cells of small emfs $\varepsilon_1 = 2V$ and $\varepsilon_2 = 4V$ having internal resistance 2Ω and 6Ω respectively, are connected as shown in the figure. If galvanometer shows no deflection at the point N, the distance of point N from the point A is equal to :



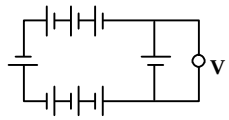
(A) $\frac{1}{6}$m (B) $\frac{1}{3}$m
(C) 25cm (D) 50cm

9. A milliameter of range 10 mA and resistance 9Ω is joined in a circuit as shown. The meter gives full-scale deflection for current I when A and B are used as its terminals, i.e., current enters at A and leaves at B (C is left isolated). The value of I is



(A) 100 mA (B) 900 mA
(C) 1 A (D) 1.1 A

10. In the circuit shown alongside, each battery is 5V and has an internal resistance of 0.2 ohm. The reading in the ideal voltmeter V is.



(A) 0 V (B) 1 V
(C) 2 V (D) 3 V

Space for Rough work

Multiple Correct Answer(s) Type

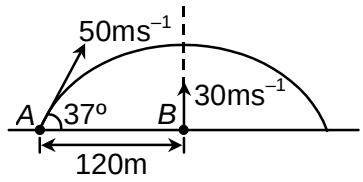
This section contains **5 multiple choice** questions. Each question has four choices (A), (B), (C) and (D) out of which **ONE or MORE THAN ONE** are correct.

11. The instantaneous velocity v of a particle is related to its displacement x according to the relation $v = ax + b$, where $a > 0$ and $b \le a/7$. Which of the following statement(s) is (are) true if $x = 0$ at $t = 0$.

(A)The displacement of the particle at time t is $x = \frac{b}{a}(e^{at} - 1)$
(B)The particle will experience a retardation if $b < 0$
(C)The particle will be at rest at time $t = 0$
(D)The acceleration of the particle is constant

12. An moving body may have
(A)Varying speed without having varying velocity
(B)Varying velocity without having varying speed
(C)Nonzero acceleration without having varying velocity
(D)Nonzero acceleration without having varying speed

13. Balls A and B are thrown from two points lying on the same horizontal plane separated by a distance 120 m. Which of the following statement(s) is(are) correct.
(A) The balls can never meet
(B) The balls can meet if the ball B is thrown 1st later
(C) The balls can meet if the balls are thrown at the same time
(D) The two balls meet at a height of 45 m



14. Which of the following statements are correct
(A) Work by kinetic function on an object can be zero
(B) Work by static friction on an object can be non zero
(C) Work by normal force on an object can be non zero
(D) Work by force which is always perpendicular to its acceleration is zero

15. A 1.5kg box is initially at rest on a horizontal surface when at $t = 0$, a horizontal force $\vec{F} = 1.8t\,\hat{i}$ N (with t in sec), is applied to the box. The acceleration of the box as a function of time t is given by

$$\begin{cases} \vec{a} = 0, \text{ for } 0 \le t \le 2.8\,\text{sec} \\ \vec{a} = (1.2t - 2.4)\hat{i} \text{ for } t > 2.8\,\text{sec} \end{cases}$$

(A) The coefficient of static friction between the box and the surface is 0.24
(B) The friction force on the box at $t = 2\,\text{sec}$ is 3.6 N
(C) The coefficient of kinetic friction between the box and surface is 0.24
(D) The friction force on the box at $t = 4\,\text{sec}$ is 3.6 N

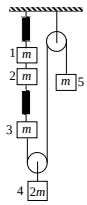
Space for Rough work

SECTION –C

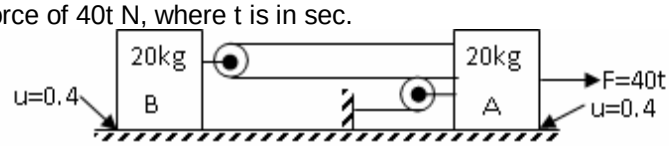
Integer Answer Type

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

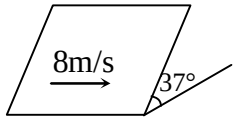
1. All the strings, springs and pulleys shown in figure are ideal. Initially the system is in equilibrium and blocks are at rest. Now, the upper spring is cut. The acceleration of block 1 just after cutting the upper spring is found y×g m/s^2. Find y?



2. Two identical blocks each having a mass of 20 kg is connected to each other by a light inextensible string as shown and is placed over a rough surface. Pulleys are connected to the blocks. The acceleration of block B after 1 sec is y/13 m/s^2 , after the application of the time varying force of 40t N, where t is in sec.

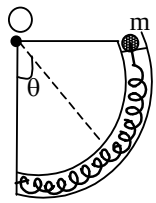


3. A ball is projected on a smooth inclined plane in a direction perpendicular to line of greatest slope with velocity of 8m/s. It's speed (in m/s) after 1 sec is 2K then value of K will be.



4. A train of length l = 350 m starts moving rectilinearly with constant acceleration $a = 3 \times 10^{-2}$ m/s^2. After 30 s the headlight of the train is switched on (event 1). After 60s from this event the tail signal light is switched on (event 2). The distance (in m) between these events in the reference frames fixed to the earth is 121K then what will be value of K.

5. A particle of mass m is released in a circular tube of radius *R*. If a spring of constant *K* is placed in the tube as shown in figure and angle made by radius vector with vertical when particle will come at instant rest is 60°, then the value of *K* is $\frac{(6mg)\lambda}{\pi^2 R}$ then find the value of λ :



Space for Rough work

Chemistry

PART – II

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions.** Each question has four choices (A), (B), (C) and (D) out of which **ONLY ONE** is correct.

1. Which of the following property is not common in both VBT as well as MOT?
   (A) LCAO
   (B) Losing of nature of atomic orbital
   (C) Mixing of atomic orbital
   (D) Wave nature of electron

2. Which of the following method is not used to convert $\underset{1^o}{ROH} \rightarrow RBr$?
   (A) $ROH \xrightarrow{HBr}$
   (B) $ROH \xrightarrow{NaBr}$
   (C) $ROH \xrightarrow[Conc.H_2SO_4]{NaBr}$
   (D) $ROH \xrightarrow[H_3PO_4]{NaBr}$

3. The Molecularity of a reaction is :
   (A) Always two
   (B) Same as its order
   (C) Different than the order
   (D) May be same or different as compared to order

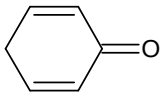
4. During micelle formation hydrocarbon part is directed towards
   (A) centre
   (B) periphery
   (C) randomly
   (D) Alternatively

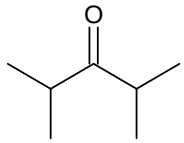
5. Which of the following is having maximum hydration energy?
   (A) V^{3+}
   (B) Fe^{2+}
   (C) Cr^{3+}
   (D) Cu^{2+}

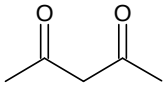
6. Which of the following is an application of Bohr's theory in an orbit?
   (A) frequency of an electron
   (B) nodal plane
   (C) no. of waves
   (D) probability of finding of electron

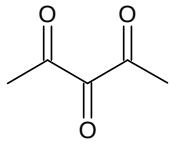
Space for Rough work

7. In which of the following ΔG_f is maximum in acidic medium?

(A) 

(B) 

(C) 

(D) 

8. $\underset{3^o}{RX} \xrightarrow{(X)} RH$

Reagent (X) cannot be used in the above conversion

(A) $LiAlH_4$ (B) $NaBH_4$

(C) Raney Ni (D) All

9. Which type of nuclear spin isomer of hydrogen exist at the temperature at which sulphonation of phenol gives major para substituted product

(A) Ortho (B) Para

(C) both (D) cant be defined

10. VBT can't explain the geometry of

(A) $\left[Cr(NH_3)_6\right]^{3+}$ (B) $\left[Co(NH_3)_6\right]^{3+}$

(C) $\left[Cu(NH_3)_4\right]^{2+}$ (D) $\left[Zn(NH_3)_4\right]^{2+}$

Space for Rough work

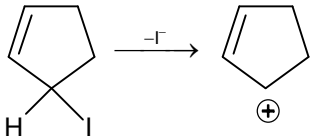
Multiple Correct Choice Type

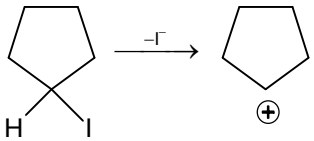
This section contains 5 multiple choice questions. Each question has 4 choices (A), (B), (C) and (D) for its answer, out which **ONE OR MORE** is/are correct.

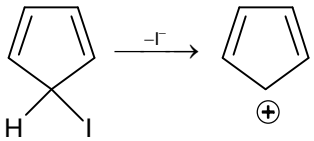
11. Which of the following ions can be separated by using NH_4Cl and NH_4OH?
(A) Fe^{3+} and Cr^{3+} (B) Cr^{3+} and Co^{2+}
(C) Cr^{3+} and Al^{3+} (D) Al^{3+} and Ba^{2+}

12. Which of the following has/have three isomers (including geometrical optical, ionisation, hydrated, linkage isomerism)?
(A) $[Zn(NH_3)(H_2O)(Cl)(Br)]$ (B) $[Cr(NH_3)_4(H_2O)_2]Br$
(C) $[Cr(NH_3)_5Cl]NO_2$ (D) $K_3[Co(C_2O_4)_3]$

13. Consider the following reactions

(I) :  (X)

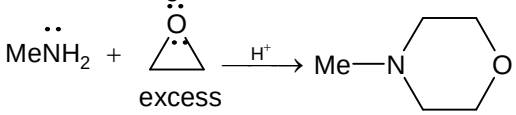
(II) :  (Y)

(III) :  (Z)

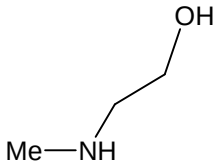
Select correct statement (s)
(A) All the carbocations (X), (Y) and (Z) are secondary
(B) X is stabilised by conjugation with the double bond and is formed fastest
(C) Z is least stable and formed slowest
(D) Z is conjugated cyclic system with four π electrons and thus is anti aromatic

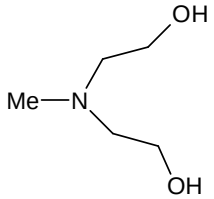
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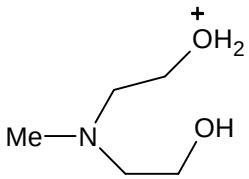
14. In the following reaction

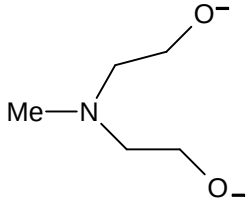


Intermediates can be

(A) 

(B) 

(C) 

(D) 

15. Which of the folllowing possess two lone pairs at central atom and square planar shape?

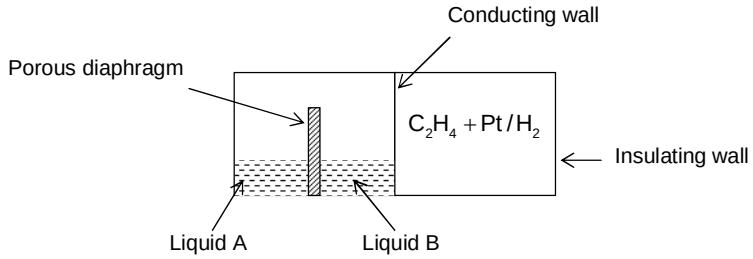
(A) SF_4 (B) XeO_4

(C) XeF_4 (D) ICl_4^-

Space for Rough work

SECTION –C
Integer Answer Type

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive).

1. 

In how many cases mixing of liquid (A) and liquid (B) is more effective if liquid (A) is H_2O and liquid (B) can be HCl, CH_3CH_2OH, HNO_3 and CH_3OCH_3

2. The electron in the ground state for a H-atom absorbs a photon of wavelength 97.20 nm and it reaches to maximum level 'n' then when it returns from this level to ground state. Maximum numbers of lines can be obtained in its spectrum?

3. The emf of the cell,
$M(s)|M_nX_n(0.01M)\ ||\ M_nX_n(0.1M)/M(s)$ is found to be 0.0295 V at 25^oC. If M_nX_n is a soluble salt, find the value of n.

4. How many complexes will show optical but not geometrical isomerism?

$\left[Pt(en)_2Cl_2\right]^{2+}$, $\left[Pt(Py)NH_3ClBr\right]$

$\left[Pt(en)(ox)Cl_2\right]$, $\left[Pt(en)_3\right]^{4+}$

5. 25 ml of 11.2 volume H_2O_2 solution is titrated with 25 ml $KMnO_4/H^+$. If 1L same $KMnO_4$ is used to oxidize 1 mole of an element X so that it is converted into a single species in which 'X' contains a single electron. If this single species is irradiated with light radiation of 114 $\mathring{A}$ so that electron is excited into a particular energy level. Determine no. of transitions possible from this energy level to ground state

Space for Rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains **10 multiple choice questions** numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which **ONLY ONE** is correct

1. The minimum value of the function f(x) = $x^{\frac{3}{2}} + x^{\frac{-3}{2}} - 4\left(x + \frac{1}{x}\right)$ for all permissible real x, is

   (A) –10 (B) –6
   (C) –7 (D) –8

2. The value of the definite integral $\int_{e}^{e^{2010}} \frac{1}{x}\left(1 + \frac{1 - \ln x}{\ln x \ln\left(\frac{x}{\ln x}\right)}\right) dx$, equals

   (A) 2009 – ln(2010 – ln 2010) (B) 2010 – ln(2009 – ln 2009)
   (C) 2009 – ln(2010 – ln 2009) (D) 2010 – ln(2010 – ln 2010)

3. Let z_1, z_2 and z_3 be three points on |z| = 1 and $z_1 + z_2 + z_3 = 0$. If θ_1, θ_2 and θ_3 be the arguments of z_1, z_2, z_3 respectively then $\cos(\theta_1 - \theta_2) + \cos(\theta_2 - \theta_3) + \cos(\theta_3 - \theta_1)$ is equal to

   (A) 0 (B) $-\frac{3}{2}$
   (C) –1 (D) $\frac{3}{2}$

4. A point (x, y) where function f(x) = [sin [x]] in $(0, 2\pi)$ is not continuous, is (where [.] denotes greatest integer $\leq$ x)

   (A) (3, 0) (B) (2, 0)
   (C) (1, 0) (D) (4, –1)

Space for rough work

5. The solution of differential equation $2x^3y\,dy + (1 - y^2)(x^2y^2 + y^2 - 1)\,dx = 0$
(A) $x^2y^2 = (cx + 1)(1 - y^2)$ (B) $x^2y^2 = (cx + 1)(1 + y^2)$
(C) $x^2y^2 = (cx - 1)(1 - y^2)$ (D) None of these

6. Let S be the array of integral points (x, y, z) with x = 0, 1, 2; y = 0, 1, 2, 3 and z = 0, 1, 2, 3, 4. If two points are chosen from S, then the probability that their mid point is in S is
(A) $\frac{4}{59}$ (B) $\frac{2}{59}$
(C) $\frac{52}{177}$ (D) None of these

7. The number of ways in which 15 boys and 2 girls can sit in a row such that between the girls at the most 2 boys sit is
(A) 17! – (12! × 3!) (B) 17! – ($^{12}C_3$ × 3!)
(C) 17! – ($^{12}C_3$ × 15!) (D) 17! – (91 × 2! × 15!)

8. If $\vec{a}$, $\vec{b}$ and $\vec{c}$ are three mutually perpendicular unit vectors and $\vec{d}$ is a unit vector which makes equal angle with $\vec{a}$, $\vec{b}$ and $\vec{c}$, then $|\vec{a} + \vec{b} + \vec{c} + \vec{d}|^2$ is equal to
(A) 4 (B) $4 \pm \sqrt{3}$
(C) $4 \pm 2\sqrt{3}$ (D) None of these

9. The line $\frac{x}{k} = \frac{y}{2} = \frac{z}{-12}$ makes an isosceles triangle with the planes 2x + y + 3z – 1 = 0 and x + 2y – 3z – 1 = 0 then value of k is
(A) 71 (B) –72
(C) 73 (D) 74

10. If $f(x) = 4x^3 - x^2 - 2x + 1$ and $g(x) = \begin{cases} \text{Min } \{f(t) : 0 \le t \le x\} & ; \ 0 \le x \le 1 \\ 3 - x & ; \ 1 < x \le 2 \end{cases}$ then $g\left(\frac{1}{4}\right) + g\left(\frac{3}{4}\right) + g\left(\frac{5}{4}\right)$ has the value equal to
(A) $\frac{7}{4}$ (B) $\frac{9}{4}$
(C) $\frac{13}{4}$ (D) $\frac{5}{2}$

Space for rough work

Multiple Correct Choice Type

This section contains **5 multiple choice questions.** Each question has 4 choices (A), (B), (C) and (D) for its answer, out of which **ONE OR MORE** is/are correct

11. The circles $x^2 + y^2 - 4x - 81 = 0$, $x^2 + y^2 + 24x - 81 = 0$ intersect each other at points A and B. A line through point A meet one circle at P and a parallel line through B meet the other circle at Q. Then the locus of the mid point of PQ is
(A) $(x + 5)^2 + (y + 0)^2 = 25$ (B) $(x - 5)^2 + (y - 0)^2 = 25$
(C) $x^2 + y^2 + 10x = 0$ (D) $x^2 + y^2 - 10x = 0$

12. Let S be the set of real values of parameter λ for which the equation $f(x) = 2x^3 - 3(2 + \lambda)x^2 + 12\lambda x$ has exactly one local maximum and exactly one local minimum. Then S is a subset of
(A) $(-4, \infty)$ (B) $(-3, 3)$
(C) $(3, \infty)$ (D) $(-\infty, 0)$

13. A and B are the points (2, 0) and (0, 2) respectively. The coordinates of the point P on the line $2x + 3y + 1 = 0$ is/are
(A) (7, –5) if |PA – PB| is maximum (B) $\left(-\frac{1}{5}, -\frac{1}{5}\right)$ if |PA – PB| is maximum
(C) (7, –5) if |PA – PB| is minimum (D) $\left(-\frac{1}{5}, -\frac{1}{5}\right)$ if |PA – PB| is minimum

14. If $x^2 - 3x + 2$ be a factor of the expression $x^4 - px^2 + q$, then
(A) roots of equation $x^4 - px^2 + q = 0$ are $\pm 1, \pm 2$
(B) roots of equation $x^4 - px^2 + q = 0$ are 1, 2, a + ib, a – ib (a, b R b ≠ 0)
(C) value of p is 5
(D) value of q is 4

15. If $f(x) = ae^{2x} + be^x + cx$ satisfies the condition $f(0) = -1$, $f'(\log 2) = 31$ and $\int_0^{\ell n 4} (f(x) - cx)\,dx = \frac{39}{2}$, then
(A) a = 5 (B) b = –6
(C) c = 2 (D) a = 3

Space for rough work

SECTION – C

(One Integer Value Correct Type)

This section contains **5 questions**. Each question, when worked out will result in **one integer** from 0 to 9 (both inclusive)

1. A straight line passing through origin O meets the lines $4x + 2y = 3$ and $2x + y + 6 = 0$ at the points P and Q respectively, then the point O divides the segment PQ in the ratio $1 : k$ then k is equal to _____

2. The number of real solutions of $\cos^{-1} x + \cos^{-1} 2x = -\pi$ is _____

3. If trace (A) > 0 and abc = 1 where $A = \begin{bmatrix} a & b & c \\ b & c & a \\ c & a & b \end{bmatrix}$ and $AA' = I$, then the value of $a^3 + b^3 + c^3$ is _____

4. Let f: $[0, \infty) \to R$ be a continuous strictly increasing function such that $f^3(x) = \int_0^x t f^2(t) dt \ \forall x \geq 0$ then value of f(6) is _____

5. The hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ passes through the point of intersection of the lines $x - 3\sqrt{5} y = 0$ and $\sqrt{5} x - 2y = 13$ and the length of its latus rectum is 4/3 units. Then the value of $9e^2 - 1$ where e is the eccentricity of the hyperbola is _____

Space for rough work